

Theoretical insights into the electronic structure and NBO features driving nonlinear optical enhancement in the donor– π –acceptor system of OPPh₃py

Ubeiden C. Samboni,^{1,2,*} Luis Felipe Berrio Velasco,¹ Fernando Cuenú Cabezas,¹ Alejandro García Ríos,¹ Christian D. Lorenz³ and Alberto Sánchez López^{1,4,*}

¹*Facultad de Ciencias Básicas y Tecnologías, Universidad del Quindío, Armenia - Quindío, Colombia*

²*Centro de Investigaciones en Química Biológica de Córdoba (CIQUIBIC), CONICET, Departamento de Química Biológica Ranwel Caputto, Facultad de Ciencias Químicas, Universidad Nacional de Córdoba, Ciudad Universitaria, Córdoba X5000HUA, Argentina*

³*Department of Engineering, King's College London, London, WC2R 2LS, United Kingdom*

⁴*Instituto de Investigaciones en Físico-Química de Córdoba (INFIQC), Departamento de Química Teórica y Computacional (DQTC), Facultad de Ciencias Químicas, Universidad Nacional de Córdoba, Ciudad Universitaria, Córdoba X5000HUA, Argentina*

E-mail: ubeiden-samboni@unc.edu.ar(UCS), alberto.sanchez@unc.edu.ar(ASL)

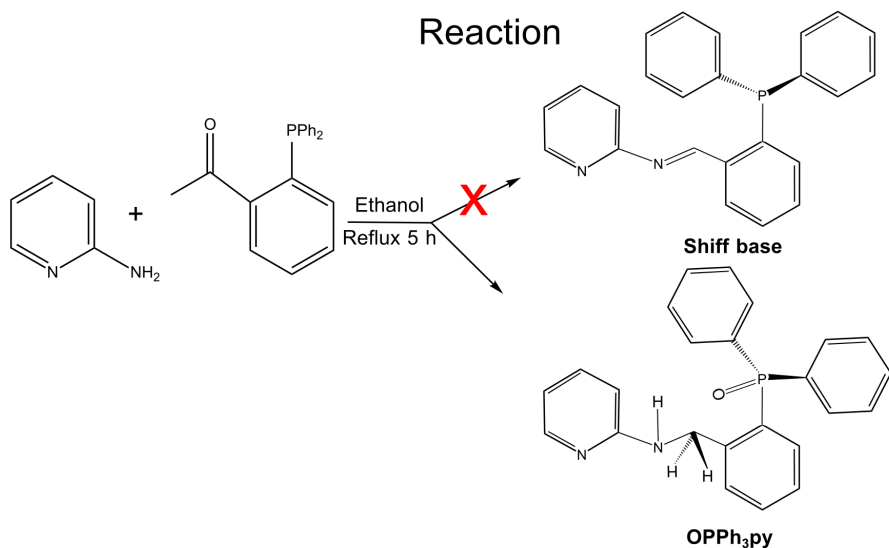
Section S1. Proposed mechanism of OPPh₃py formation

2-(diphenylphosphinyl)benzaldehyde reacts with primary amines to generate hemilabile ligands. In the resulting condensation reaction, the aldehyde group typically forms imines (Schiff bases).¹ However, factors such as temperature, solvent choice, and acid catalysis can steer the reaction toward the formation of aminophosphines. Previous studies have documented these transformations under various conditions^{2,3}.

In this study, the reaction of 2-(diphenylphosphinyl)benzaldehyde with 2-aminopyridine was carried out in ethanol under reflux; instead of the expected Schiff base, the OPPh₃py product was isolated (Scheme S1A). A mechanism is proposed as follows (Scheme S1B). First, the primary amine attacks the carbonyl group, forming a dipolar intermediate. Second, proton transfer generates a carbinolamine intermediate. Next, the oxygen atom interacts with the phosphorus center due to their proximity. This interaction promotes intramolecular transfer from oxygen to phosphorus. A concurrent proton shift occurs from the hydroxyl group to the adjacent carbon. Finally, the positively charged phosphorus species undergoes resonance stabilisation. This process leads to the formation of a phosphorus-oxygen double bond. This mechanism is consistent with that reported by Popov et al.⁴

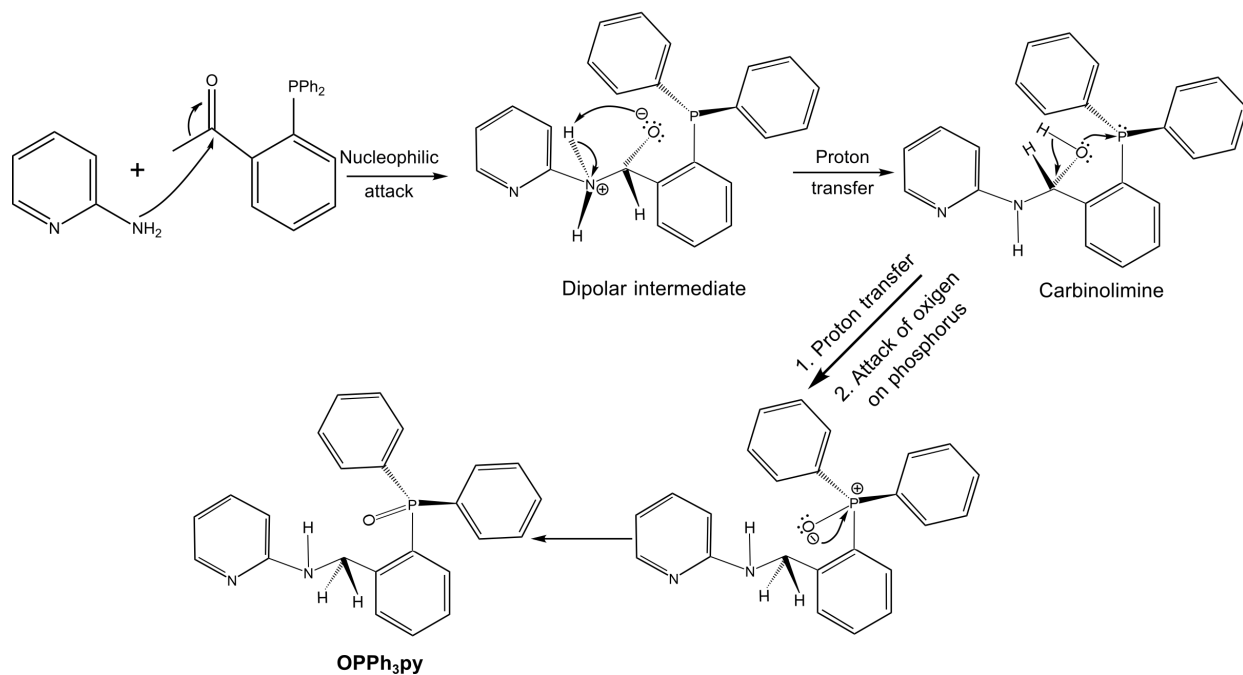
(A)

Reaction



(B)

Mechanism proposed



Scheme S1: Synthesis of OPPh₃py in ethanol and its mechanism. (A) Under these conditions, the reaction yields a single product containing both primary amine and phosphine oxide groups, rather than a Schiff base. (B) Proposed mechanism for OPPh₃py formation.

Table S1: Optimised bond lengths (Å) for OPPh₃py at the DFT B3LYP-D3(BJ)/6-311++G(2df,p) level of theory.

Bond (Å)	^a Experimental		Optimised	Deviation (Exp _{avg} – Calc.)
	Isomer I	Isomer II		
P1–O2	1.49 (17)	1.48 (18)	1.51	0.03
P1–C3	1.80 (2)	1.81 (3)	1.83	0.03
P1–C14	1.80 (3)	1.80 (3)	1.83	0.03
P1–C25	1.81 (2)	1.81 (3)	1.83	0.02
C3–C4	1.36 (3)	1.38 (4)	1.40	0.02
C3–C12	1.38 (4)	1.38 (4)	1.40	0.02
C4–C6	1.38 (4)	1.39 (4)	1.39	0.01
C6–C8	1.36 (4)	1.36 (4)	1.40	0.03
C8–C10	1.34 (4)	1.36 (5)	1.40	0.04
C10–C12	1.37 (4)	1.38 (5)	1.39	0.02
C14–C23	1.39 (4)	1.38 (4)	1.40	0.01
C14–C15	1.39 (3)	1.38 (3)	1.40	0.01
C15–C17	1.37 (4)	1.38 (4)	1.39	0.02
C17–C19	1.37 (5)	1.36 (4)	1.39	0.02
C19–C21	1.36 (4)	1.37 (4)	1.40	0.02
C21–C23	1.39 (4)	1.38 (4)	1.39	0.01
C25–C33	1.39 (3)	1.39 (3)	1.40	0.01
C25–C26	1.42 (3)	1.40 (3)	1.41	0.01
C26–C27	1.39 (4)	1.38 (4)	1.40	0.01
C26–C35	1.52 (4)	1.52 (4)	1.52	0.01
C27–C29	1.37 (4)	1.38 (4)	1.39	0.02
C29–C31	1.36 (4)	1.36 (4)	1.39	0.03
C31–C33	1.38 (3)	1.38 (4)	1.39	0.02
C35–N38	1.43 (3)	1.44 (3)	1.46	0.03
N38–C40	1.35 (4)	1.36 (3)	1.37	0.02
C40–N41	1.34 (3)	1.33 (3)	1.34	0.01
C40–C48	1.39 (4)	1.40 (4)	1.42	0.02
N41–C42	1.34 (4)	1.35 (4)	1.34	0.01
C42–C44	1.35 (5)	1.35 (5)	1.39	0.04
C44–C46	1.37 (6)	1.37 (5)	1.40	0.03
C46–C48	1.36 (5)	1.35 (4)	1.38	0.03

^aNumbers in parentheses represent estimated standard uncertainties in the last digit(s) of the value.

Table S2: Angles (°) for OPPh₃py at the DFT B3LYP-D3(BJ)/6-311++G(2df,p)) level of theory.

Angle (°)	^a Experimental		Optimised	Deviation (Exp _{avg} - Calc.)
	Isomer I	Isomer II		
O2-P1-C3	112.39 (11)	112.62 (12)	112.33	0.16
O2-P1-C14	110.16 (11)	111.44 (12)	111.29	0.79
C3-P1-C14	106.34 (12)	106.42 (12)	106.04	0.34
O2-P1-C25	112.55 (11)	114.12 (11)	114.13	1.73
C3-P1-C25	106.98 (11)	104.94 (11)	105.43	1.23
C14-P1-C25	108.14 (12)	106.73 (12)	107.07	0.99
C4-C3-C12	117.70 (20)	118.00 (30)	119.88	1.99
C4-C3-P1	124.10 (20)	118.30 (20)	122.51	1.90
C12-C3-P1	118.00 (20)	123.60 (20)	117.61	2.00
C3-C4-C6	120.90 (20)	120.50 (30)	119.97	0.83
C8-C6-C4	119.80 (30)	120.50 (30)	120.02	0.36
C10-C8-C6	120.20 (30)	119.70 (30)	120.09	0.11
C8-C10-C12	120.20 (30)	120.50 (30)	120.09	0.31
C10-C12-C3	121.20 (30)	120.80 (30)	119.94	0.93
C23-C14-C15	118.80 (30)	118.40 (30)	119.83	0.81
C23-C14-P1	123.30 (20)	119.40 (20)	123.08	1.23
C15-C14-P1	117.90 (20)	122.30 (20)	117.09	1.06
C17-C15-C14	120.40 (30)	120.30 (30)	119.97	0.33
C15-C17-C19	120.30 (30)	120.20 (30)	120.09	0.21
C21-C19-C17	120.20 (30)	120.50 (30)	120.06	0.27
C19-C21-C23	120.30 (30)	119.40 (30)	120.05	0.17
C21-C23-C14	120.00 (30)	121.20 (30)	120.00	0.20
C33-C25-C26	118.80 (20)	119.20 (20)	119.85	0.77
C33-C25-P1	120.60 (20)	120.30 (19)	120.19	0.36
C26-C25-P1	120.59 (19)	120.45 (19)	119.96	0.54
C27-C26-C25	118.00 (20)	118.20 (20)	118.46	0.33
C27-C26-C35	120.20 (20)	121.40 (20)	118.82	1.39
C25-C26-C35	121.70 (20)	120.40 (20)	122.66	1.38
C29-C27-C26	121.50 (30)	121.80 (30)	121.35	0.28
C31-C29-C27	120.80 (30)	120.00 (30)	120.01	0.39
C29-C31-C33	119.10 (30)	119.40 (30)	119.60	0.50
C31-C33-C25	121.70 (30)	121.40 (30)	120.73	0.83

Table S2: (Continue)

Angle (°)	^a Experimental		Optimised	Deviation (Exp _{avg} - Calc.)
	Isomer I	Isomer II		
N38–C35–C26	115.50 (20)	115.40 (20)	113.20	2.15
C40–N38–C35	122.20 (30)	122.90 (20)	121.75	0.83
N41–C40–N38	117.70 (30)	118.10 (30)	118.29	0.39
N41–C40–C48	121.60 (30)	122.90 (30)	121.99	0.19
N38–C40–C48	120.70 (20)	119.10 (20)	119.71	1.00
C40–N41–C42	117.30 (30)	115.90 (30)	117.99	0.55
N41–C42–C44	124.80 (40)	124.60 (30)	124.34	0.41
C42–C44–C46	117.40 (40)	118.40 (30)	117.40	0.00
C48–C46–C44	120.60 (40)	119.50 (30)	119.57	1.03
C46–C48–C40	118.40 (40)	118.80 (30)	118.70	0.25

^a Numbers in parentheses represent estimated standard uncertainties in the last digit(s) of the value.

The following molecules ([Table S3](#)) were selected to benchmark the global reactivity parameters of OPPh₃py. These molecules were chosen because of their similarity in structural formula, each having triphenylphosphine oxide as the base structural unit. Although the reference molecules selected share a common structural framework with OPPh₃py, their reported reactivity parameters were not obtained at the same level of theory used in our calculations (B3LYP-D3(BJ)/6-311G++(2df,p)). As a result, the comparison presented here should be regarded as qualitative rather than quantitative. Nonetheless, structural similarity among the chosen molecules provides a reasonable basis for establishing general trends. To strengthen the validity of our analysis, all comparisons are interpreted cautiously, and the limitations arising from cross-method data are explicitly acknowledged.

Table S3: Molecular structures of the reference compounds used for benchmarking the global reactivity parameters of OPPh_3py .

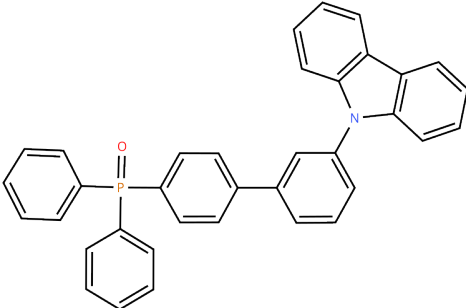
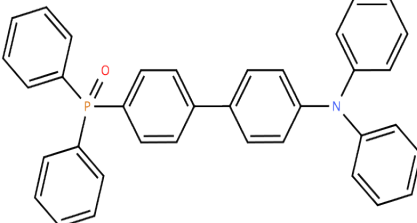
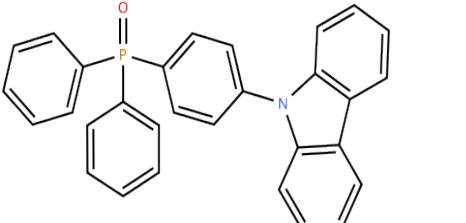
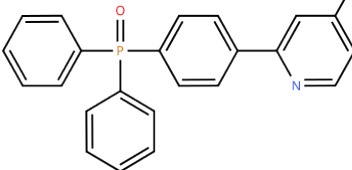
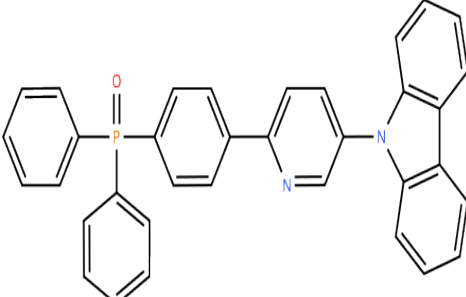
Molecule	Rename	Structural Formula
1 Ref. ⁵	TPP1	
POA Ref. ⁶	TPP2	
MPO12 Ref. ⁷	TPP3	
pPO-ppy-mCZ Ref. ⁸	TPP4	
pPO-ppy-pCZ Ref. ⁸	TPP5	

Table S4: Local reactivity parameters: Fukui functions using CM5 population analysis: local softness ($s_k^\pm$) and philicity ($\omega_k^\pm$) in eV for selected atomic sites of OPPh₃py. Values are scaled by 10^{-2} for $f_k^\pm$ and $s_k^\pm$, and by 10^{-1} for $\omega_k^\pm$.

Atom	Fukui function		Local softness		Philicity	
	f_k^+	f_k^-	s_k^+	s_k^-	ω_k^+	ω_k^-
P1	3.64	0.09	1.04	0.03	6.71	0.17
O2	2.37	0.01	0.68	0.00	4.37	0.02
C3	2.53	-0.70	0.72	-0.20	4.67	-1.29
C4	3.08	-0.13	0.88	-0.04	5.68	-0.24
C6	2.71	0.90	0.78	0.26	5.00	1.66
C8	5.58	1.46	1.60	0.42	10.29	2.69
C10	4.00	0.98	1.15	0.28	7.38	1.80
C12	2.28	-0.07	0.65	-0.02	4.20	-0.13
C14	2.61	-0.54	0.75	-0.15	4.81	-0.99
C15	3.60	-0.18	1.03	-0.05	6.64	-0.32
C17	3.08	0.83	0.88	0.24	5.68	1.53
C19	5.74	1.41	1.64	0.40	10.59	2.60
C21	3.64	1.06	1.04	0.30	6.71	1.96
C23	1.87	0.21	0.54	0.06	3.45	0.39
C25	1.73	0.54	0.50	0.15	3.19	0.99
C26	1.61	-0.66	0.46	-0.19	2.97	-1.22
C27	2.13	0.33	0.61	0.09	3.93	0.61
C29	4.27	1.55	1.22	0.44	7.87	2.86
C31	2.47	2.76	0.71	0.79	4.56	5.09
C33	1.39	1.72	0.40	0.49	2.56	3.17
C35	0.45	2.16	0.13	0.62	0.83	3.98
N38	-0.17	15.34	-0.05	4.39	-0.31	28.29
C40	-0.06	3.34	-0.02	0.96	-0.11	6.16
N41	0.23	5.75	0.07	1.65	0.43	10.60
C42	1.28	6.09	0.37	1.74	2.36	11.23
C44	1.78	10.46	0.51	2.99	3.29	19.29
C46	1.78	4.53	0.51	1.30	3.29	8.36
C48	0.64	7.76	0.18	2.22	1.18	14.32

Table S5: Second order perturbation theory analysis of the Fock matrix in the NBO basis for OPPh₃py.

Donor (<i>i</i>)	Type	ED(<i>i</i>) (e)	Acceptor (<i>j</i>)	ED(<i>j</i>) (e)	Type	^a <i>E</i> (2) (kcal/mol)	^b <i>E</i> (<i>j</i>) − <i>E</i> (<i>i</i>) (a.u)	^c <i>F</i> (<i>i</i> , <i>j</i>) (a.u)
LP O2	-	1.976	BD* N38-H39	0.028	σ*	2.94	1.03	0.049
LP O2	-	1.976	RY P1	0.009	-	7.03	1.76	0.099
LP O2	-	1.833	BD* P1-C14	0.118	π*	17.75	0.5	0.084
LP O2	-	1.833	RY P1	0.029	-	8.36	1.26	0.092
LP O2	-	1.832	BD* P1-C3	0.125	π*	17.96	0.49	0.084
LP O2	-	1.832	BD* P1-C25	0.123	π*	9.49	0.49	0.061
LP O2	-	1.832	RY* P1	0.030	-	8.54	1.23	0.091
LP N38	-	1.743	BD* C26-C35	0.035	π*	8.2	0.64	0.065
LP N38	-	1.743	BD* C40-N41	0.460	π*	53.84	0.26	0.105
LP N41	-	1.909	BD* C40-C48	0.032	π*	9.68	0.83	0.08
LP N41	-	1.909	BD* C42-C44	0.243	π*	8.53	0.87	0.077
BD C3-C4	π	1.681	BD* C6-C8	0.322	π*	18.13	0.29	0.065
BD C3-C12	π	1.681	BD* C10-C12	0.303	π*	20.33	0.29	0.069
BD C6-C8	π	1.650	BD* C3-C4	0.377	π*	23.02	0.27	0.071
BD C6-C8	π	1.650	BD* C10-C12	0.303	π*	19.32	0.28	0.066
BD C10-C12	π	1.642	BD* C3-C4	0.377	π*	20.34	0.27	0.066
BD C10-C12	π	1.642	BD* C6-C8	0.322	π*	21.79	0.28	0.069
BD C14-C15	π	1.671	BD* C17-C19	0.319	π*	17.98	0.29	0.064
BD C14-C15	π	1.671	BD* C21-C23	0.315	π*	21.37	0.28	0.07
BD C17-C19	π	1.650	BD* C14-C15	0.360	π*	23.08	0.28	0.071
BD C17-C19	π	1.650	BD* C21-C23	0.315	π*	19.85	0.28	0.066
BD C21-C23	π	1.657	BD* C14-C15	0.360	π*	19.27	0.28	0.065
BD C21-C23	π	1.657	BD* C17-C19	0.319	π*	20.91	0.28	0.069
BD C25-C33	π	1.694	BD* C26-C27	0.305	π*	19.88	0.3	0.069
BD C25-C33	π	1.694	BD* C29-C31	0.320	π*	17.25	0.29	0.063
BD C26-C27	π	1.629	BD* C25-C33	0.373	π*	20.03	0.26	0.065
BD C26-C27	π	1.629	BD* C29-C31	0.320	π*	22.31	0.27	0.07
BD C29-C31	π	1.656	BD* C25-C33	0.373	π*	23.6	0.27	0.072
BD C29-C31	π	1.656	BD* C26-C27	0.305	π*	18.8	0.29	0.066
BD C40-N41	π	1.710	BD* C42-C44	0.311	π*	30.15	0.32	0.088
BD C42-C44	π	1.697	BD* C40-N41	0.460	π*	13.34	0.26	0.053
BD C42-C44	π	1.697	BD* C46-C48	0.288	π*	24.22	0.28	0.074
BD C46-C48	π	1.732	BD* C40-N41	0.460	π*	28.86	0.27	0.079
BD C46-C48	π	1.732	BD* C42-C44	0.311	π*	14.35	0.29	0.057

^a *E*(2) means energy of hyper conjugative interactions. ^b Energy difference between donor and acceptor *i* and *j* NBO orbitals

^c *F*(*i*, *j*) is the Fock matrix element between *i* and *j* NBO orbitals

Nonlinear optical (NLO) properties reported in the following supporting tables were calculated using the levels of theory summarised below:

Method	Empirical Dispersion	Basis Set
B3LYP	GD3(BJ)	6-311++G(2df,p)
CAM-B3LYP	GD3(BJ)	6-311++G(2df,p)
M06-2X	GD3	6-311++G(2df,p)
*M06-2X	GD3	def2-TZVPPD
ω B97XD	–	6-311++G(2df,p)

All properties were calculated using different solvation models: gas phase (GAS), polarisable continuum model (PCM) and solvation model based on density(SMD) and at multiple wavelengths ($\lambda = \infty$, 532, 625, 1064, 1341, and 1906 nm). However, values corresponding to 532, and 625 nm are not reported in the main supporting tables due to the large discrepancies and irregular behaviour observed in these data. The complete raw dataset is available at: <https://doi.org/10.5281/zenodo.18866407>.

Table S6: Calculated dipole moment (μ_D), isotropic polarisability ($\alpha_{\text{iso}} \times 10^{-24}$ esu), and anisotropic polarisability ($\alpha_{\text{aniso}} \times 10^{-24}$ esu) of OPPh₃py at different wavelengths and levels of theory.

Model	λ	μ_D					α_{iso}					α_{aniso}				
		B3LYP	CAM-B3LYP	M06-2X	*M06-2X	ω B97XD	B3LYP	CAM-B3LYP	M06-2X	*M06-2X	ω B97XD	B3LYP	CAM-B3LYP	M06-2X	*M06-2X	ω B97XD
GAS	1064	4.80	4.76	4.57	4.58	4.67	48.25	46.84	46.83	47.31	47.01	11.26	10.54	10.74	10.85	10.63
	1341	4.80	4.76	4.57	4.58	4.67	47.97	46.59	46.58	47.07	46.76	11.11	10.41	10.61	10.72	10.50
	1906	4.80	4.76	4.57	4.58	4.67	47.73	46.38	46.37	46.86	46.55	10.98	10.31	10.50	10.61	10.40
	∞	4.80	4.76	4.57	4.58	4.67	47.51	46.18	46.17	46.65	46.34	10.86	10.21	10.39	10.51	10.29
PCM	1064	6.13	6.07	5.80	5.83	5.94	54.82	53.10	53.11	53.74	53.34	11.63	10.85	11.02	11.14	10.95
	1341	6.13	6.07	5.80	5.83	5.94	54.47	52.79	52.80	53.42	53.02	11.45	10.71	10.88	10.99	10.80
	1906	6.13	6.07	5.80	5.83	5.94	54.17	52.52	52.53	53.15	52.75	11.31	10.59	10.75	10.86	10.68
	∞	6.13	6.07	5.80	5.83	5.94	59.45	57.54	57.57	58.35	57.85	10.42	9.76	9.88	9.97	9.81
SMD	1064	6.33	6.27	5.99	6.04	6.13	56.23	54.46	54.49	55.14	54.71	11.75	10.94	11.35	11.49	10.98
	1341	6.33	6.27	5.99	6.04	6.13	55.86	54.13	54.16	54.81	54.38	11.57	10.79	11.19	11.34	10.83
	1906	6.33	6.27	5.99	6.04	6.13	55.54	53.85	53.88	54.53	54.10	11.42	10.67	11.06	11.21	10.70
	∞	6.33	6.27	5.99	6.04	6.13	62.19	60.18	60.26	61.11	60.55	10.55	9.84	10.22	10.35	9.84

*Calculations for this functional were performed using the def2-TZVPPD basis set.

Table S7: Selected components contributing tensors and values of the static and frequency dependant first-order hyperpolarisability, β (static ($\beta(0; 0, 0) \times 10^{-30}$ esu, dynamic ($\beta^\lambda(-\omega; \omega, 0) \times 10^{-30}$ esu) of OPPh₃py calculated at different levels of theory. All calculations employed the 6-311++G(2df,p) basis set unless otherwise indicated.

Model	Method	λ (nm)	β_{xxx}	β_{yyy}	β_{zzz}	β_x	β_y	β_z	$\beta_{ }$	β_{tot}
GAS	B3LYP	1064	0.53	-0.47	1.39	4.66	7.79	3.01	1.91	9.56
		1341	0.53	-0.46	1.21	4.57	7.59	2.51	1.84	9.21
		1906	0.52	-0.46	1.08	4.50	7.43	2.14	1.79	8.94
		∞	0.35	-0.73	-2.77	5.03	-5.59	-4.39	1.74	8.71
	CAM-B3LYP	1064	0.71	-0.47	-1.03	5.21	7.32	-5.04	2.06	10.30
		1341	0.70	-0.46	-1.01	5.07	7.14	-4.93	2.01	10.05
		1906	0.68	-0.46	-1.00	4.96	6.99	-4.83	1.97	9.84
		∞	0.45	-1.08	-1.29	4.21	-8.53	1.49	1.93	9.63
	M06-2X	1064	0.62	-0.40	-0.65	4.97	7.53	-4.44	2.01	10.06
		1341	0.61	-0.40	-0.64	4.83	7.34	-4.34	1.96	9.80
		1906	0.59	-0.40	-0.64	4.71	7.19	-4.25	1.92	9.59
		∞	0.34	-1.12	-1.35	4.02	-8.41	1.16	1.88	9.39
	M06-2X/ def2-TZVPPD	1064	0.55	-0.36	-0.59	5.03	6.96	-4.37	1.93	9.63
		1341	0.54	-0.35	-0.59	4.90	6.80	-4.29	1.88	9.42
		1906	0.53	-0.35	-0.59	4.80	6.67	-4.21	1.85	9.23
		∞	0.28	-1.19	-1.26	4.11	-7.97	1.29	1.81	9.06
	ω B97XD	1064	0.61	-0.47	-0.96	4.67	7.30	-5.09	2.01	10.05
		1341	0.60	-0.46	-0.94	4.53	7.12	-4.97	1.96	9.80
		1906	0.58	-0.46	-0.93	4.42	6.98	-4.87	1.92	9.59
		∞	0.34	-1.13	-1.30	3.66	-8.52	1.50	1.88	9.40
PCM	B3LYP	1064	0.97	-1.07	0.64	7.64	11.56	-2.97	2.83	14.17
		1341	0.95	-1.06	0.50	7.43	11.22	-3.22	2.77	13.84
		1906	0.93	-1.04	0.40	7.26	10.95	-3.41	2.71	13.57
		∞	0.50	-1.54	-3.70	7.94	-13.12	-1.16	3.08	15.38
	CAM-B3LYP	1064	1.18	-1.02	-1.73	8.15	10.62	-10.42	3.39	16.97
		1341	1.15	-1.01	-1.68	7.91	10.33	-10.13	3.30	16.49
		1906	1.13	-1.00	-1.65	7.70	10.10	-9.88	3.22	16.09
		∞	0.70	-1.78	-1.78	6.77	-15.40	5.59	3.55	17.73
	M06-2X	1064	1.07	-0.84	-1.06	7.82	11.19	-9.36	3.31	16.55
		1341	1.04	-0.84	-1.04	7.57	10.88	-9.10	3.22	16.08
		1906	1.02	-0.83	-1.02	7.37	10.63	-8.88	3.14	15.69
		∞	0.50	-1.92	-1.94	6.44	-15.28	4.97	3.46	17.31
	M06-2X/ def2-TZVPPD	1064	1.05	-0.77	-1.04	8.03	10.64	-9.57	3.28	16.41
		1341	1.02	-0.76	-1.03	7.80	10.37	-9.32	3.19	15.97
		1906	1.00	-0.75	-1.01	7.60	10.14	-9.11	3.12	15.61
		∞	0.46	-2.06	-1.79	6.71	-15.10	5.41	3.48	17.39
	ω B97XD	1064	1.15	-1.02	-1.60	7.77	10.64	-10.39	3.36	16.78
		1341	1.12	-1.01	-1.56	7.53	10.36	-10.10	3.26	16.31
		1906	1.10	-0.99	-1.53	7.33	10.14	-9.86	3.19	15.93
		∞	0.64	-1.87	-1.82	6.30	-15.40	5.62	3.51	17.57

Table S7: (Continue)

Model	Method	λ (nm)	β_{xxx}	β_{yyy}	β_{zzz}	β_x	β_y	β_z	$\beta_{ }$	β_{tot}
SMD	B3LYP	1064	1.15	-1.43	-0.26	8.79	11.43	-5.68	3.10	15.50
		1341	1.12	-1.41	-0.36	8.54	11.08	-5.82	3.03	15.16
		1906	1.10	-1.39	-0.44	8.34	10.80	-5.92	2.97	14.87
		∞	0.61	-1.38	-3.43	8.91	-14.52	1.64	3.42	17.12
	CAM-B3LYP	1064	1.32	-1.38	-2.52	9.07	10.21	-12.74	3.73	18.67
		1341	1.28	-1.35	-2.45	8.79	9.93	-12.37	3.63	18.13
		1906	1.26	-1.34	-2.40	8.56	9.70	-12.06	3.54	17.69
		∞	0.79	-1.61	-1.50	7.56	-16.49	8.16	3.98	19.89
	M06-2X	1064	1.22	-1.22	-1.76	8.85	10.98	-11.63	3.66	18.28
		1341	1.19	-1.20	-1.71	8.57	10.68	-11.29	3.55	17.74
		1906	1.16	-1.19	-1.68	8.34	10.43	-11.01	3.46	17.30
		∞	0.56	-1.79	-1.72	7.27	-16.53	7.40	3.90	19.51
	M06-2X/ def2-TZVPPD	1064	1.21	-1.12	-1.75	9.23	10.45	-11.89	3.67	18.33
		1341	1.18	-1.10	-1.71	8.95	10.18	-11.56	3.56	17.82
		1906	1.16	-1.08	-1.68	8.73	9.96	-11.29	3.48	17.40
		∞	0.55	-1.96	-1.54	7.79	-16.47	7.95	3.97	19.87
	ω B97XD	1064	1.29	-1.43	-2.33	8.74	9.76	-12.37	3.60	18.02
		1341	1.26	-1.41	-2.27	8.47	9.49	-12.02	3.50	17.50
		1906	1.23	-1.39	-2.22	8.25	9.27	-11.72	3.41	17.07
		∞	0.73	-1.54	-1.48	7.17	-15.88	8.20	3.85	19.25

Table S8: Selected components contributing tensors and values of the second-harmonics generation, $\beta_{\text{SHG}} (\beta^\lambda(-2\omega; \omega, \omega) \times 10^{-30} \text{ esu})$ of OPPh₃py calculated at different levels of theory. All calculations employed the 6-311++G(2df,p) basis set unless otherwise indicated.

Model	Method	λ (nm)	β_{xxx}	β_{yyy}	β_{zzz}	β_x	β_y	β_z	$\beta_{ }$	β_{SHG}
GAS	B3LYP	1064	0.57	-0.51	2.97	5.05	8.92	7.67	2.56	12.80
		1341	0.55	-0.49	1.93	4.86	8.28	4.57	2.13	10.64
		1906	0.53	-0.47	1.36	4.65	7.75	2.92	1.90	9.50
	CAM-B3LYP	1064	0.81	-0.49	-1.10	6.10	8.45	-5.71	2.38	11.88
		1341	0.75	-0.48	-1.06	5.58	7.79	-5.33	2.19	10.96
		1906	0.71	-0.47	-1.03	5.19	7.29	-5.02	2.05	10.26
	M06-2X	1064	0.71	-0.42	-0.65	5.86	8.70	-4.98	2.32	11.61
		1341	0.66	-0.41	-0.66	5.34	8.01	-4.68	2.14	10.71
		1906	0.62	-0.40	-0.65	4.95	7.50	-4.42	2.00	10.01
	M06-2X/ def2-TZVPPD	1064	0.63	-0.39	-0.55	5.84	7.92	-4.79	2.19	10.94
		1341	0.58	-0.37	-0.58	5.37	7.36	-4.57	2.04	10.19
		1906	0.55	-0.36	-0.59	5.01	6.93	-4.36	1.92	9.60
	ω B97XD	1064	0.70	-0.49	-1.03	5.51	8.38	-5.81	2.32	11.59
		1341	0.65	-0.48	-0.99	5.01	7.74	-5.40	2.14	10.69
		1906	0.61	-0.47	-0.95	4.64	7.27	-5.07	2.00	10.00
PCM	B3LYP	1064	0.97	-1.00	1.61	8.20	12.55	-0.45	3.00	14.99
		1341	0.92	-0.97	0.85	7.57	11.43	-2.23	2.78	13.89
		1906	0.87	-0.94	0.43	7.06	10.58	-3.09	2.62	13.09
	CAM-B3LYP	1064	1.24	-0.95	-1.99	9.06	11.55	-11.74	3.76	18.80
		1341	1.14	-0.92	-1.85	8.20	10.55	-10.73	3.43	17.13
		1906	1.07	-0.89	-1.74	7.56	9.81	-9.95	3.18	15.89
	M06-2X	1064	1.14	-0.76	-1.24	8.77	12.21	-10.50	3.67	18.34
		1341	1.04	-0.74	-1.18	7.90	11.14	-9.62	3.34	16.70
		1906	0.97	-0.73	-1.11	7.26	10.35	-8.93	3.10	15.48
	M06-2X/ def2-TZVPPD	1064	1.09	-0.71	-1.19	8.84	11.42	-10.56	3.58	17.89
		1341	1.00	-0.68	-1.14	8.03	10.50	-9.73	3.28	16.41
		1906	0.94	-0.66	-1.09	7.42	9.81	-9.08	3.06	15.29
	ω B97XD	1064	1.21	-0.94	-1.86	8.65	11.52	-11.69	3.71	18.56
		1341	1.11	-0.91	-1.73	7.82	10.55	-10.67	3.38	16.92
		1906	1.04	-0.89	-1.62	7.20	9.83	-9.90	3.14	15.70

Table S8: (Continue)

Model	Method	λ (nm)	β_{xxx}	β_{yyy}	β_{zzz}	β_x	β_y	β_z	$\beta_{ }$	β_{SHG}
SMD	B3LYP	1064	1.15	-1.33	0.56	9.33	12.54	-3.60	3.21	16.04
		1341	1.08	-1.27	-0.05	8.56	11.34	-4.91	3.01	15.03
		1906	1.02	-1.22	-0.37	7.96	10.44	-5.46	2.84	14.22
	CAM-B3LYP	1064	1.36	-1.27	-2.83	9.92	11.12	-14.18	4.11	20.57
		1341	1.26	-1.22	-2.61	8.96	10.14	-12.90	3.74	18.69
		1906	1.18	-1.18	-2.44	8.25	9.42	-11.93	3.46	17.30
	M06-2X	1064	1.29	-1.10	-1.99	9.79	11.99	-12.89	4.03	20.14
		1341	1.18	-1.07	-1.85	8.80	10.92	-11.73	3.66	18.29
		1906	1.10	-1.03	-1.73	8.07	10.13	-10.86	3.38	16.90
	M06-2X/ def2-TZVPPD	1064	1.24	-1.03	-1.94	9.98	11.23	-12.99	3.97	19.86
		1341	1.15	-0.98	-1.81	9.05	10.29	-11.89	3.63	18.14
		1906	1.07	-0.94	-1.71	8.36	9.60	-11.05	3.37	16.85
	ω B97XD	1064	1.34	-1.33	-2.62	9.55	10.63	-13.73	3.96	19.82
		1341	1.23	-1.27	-2.41	8.62	9.69	-12.49	3.60	18.01
		1906	1.15	-1.23	-2.25	7.94	9.00	-11.56	3.33	16.66

Table S9: Selected components contributing tensors and values of the static and frequency dependant second-order hyperpolarisability, γ (static $\gamma(0; 0, 0, 0) \times 10^{-36}$ esu, dynamic $\gamma^\lambda(-\omega; \omega, 0, 0) \times 10^{-36}$ esu) of OPPh₃py calculated at different levels of theory. All calculations employed the 6-311++G(2df,p) basis set unless otherwise indicated.

Model	Method	λ	γ_{xxyy}	γ_{yyzz}	γ_{xxzz}	γ_{xxxx}	γ_{yyyy}	γ_{zzzz}	γ
GAS	B3LYP	1064	14.94	18.82	15.71	47.98	40.15	106.44	58.62
		1341	14.49	18.09	15.17	46.62	39.10	100.50	56.30
		1906	14.13	17.52	14.74	45.51	38.24	95.91	54.47
		∞	17.32	23.42	11.15	44.18	58.96	57.04	52.79
	CAM-B3LYP	1064	11.22	12.86	11.54	35.75	30.77	60.81	39.66
		1341	10.96	12.52	11.26	34.94	30.12	58.82	38.64
		1906	10.74	12.24	11.03	34.27	29.58	57.21	37.80
		∞	12.81	15.36	8.45	33.47	41.52	36.80	37.01
	M06-2X	1064	10.97	12.58	11.15	34.78	29.64	58.91	38.49
		1341	10.70	12.23	10.87	33.97	28.99	56.91	37.46
		1906	10.47	11.95	10.63	33.29	28.45	55.29	36.62
		∞	12.42	15.00	8.03	32.38	40.52	35.33	35.82
	M06-2X/ def2-TZVPPD	1064	9.45	11.32	10.31	30.95	26.21	53.62	34.53
		1341	9.23	11.03	10.06	30.28	25.69	51.85	33.65
		1906	9.05	10.79	9.85	29.72	25.26	50.41	32.94
		∞	11.02	13.56	7.13	28.66	37.16	32.08	32.26
	ω B97XD	1064	10.84	12.23	10.96	34.27	30.12	58.08	38.06
		1341	10.60	11.92	10.71	33.53	29.51	56.28	37.13
		1906	10.39	11.66	10.50	32.92	29.00	54.82	36.36
		∞	12.29	14.69	8.28	32.26	39.75	35.64	35.63
PCM	B3LYP	1064	25.36	28.38	23.80	78.63	69.76	149.35	90.89
		1341	24.58	27.32	23.02	76.38	67.88	141.78	87.56
		1906	23.95	26.48	22.39	74.54	66.34	135.86	84.91
		∞	32.60	38.72	21.11	85.73	104.13	103.23	95.59
	CAM-B3LYP	1064	18.93	19.77	17.80	58.07	52.65	88.56	62.70
		1341	18.46	19.22	17.35	56.72	51.49	85.73	61.08
		1906	18.08	18.78	16.98	55.60	50.53	83.43	59.75
		∞	24.00	26.60	15.92	63.90	73.93	67.46	67.67
	M06-2X	1064	18.79	19.72	17.36	57.27	51.48	86.02	61.52
		1341	18.31	19.15	16.91	55.90	50.30	83.17	59.88
		1906	17.92	18.69	16.53	54.77	49.33	80.86	58.54
		∞	23.61	26.20	15.38	62.90	73.48	65.36	66.43
	M06-2X/ def2-TZVPPD	1064	17.14	18.65	16.75	53.68	48.66	81.29	58.00
		1341	16.72	18.14	16.33	52.47	47.63	78.67	56.53
		1906	16.38	17.72	15.98	51.46	46.78	76.54	55.33
		∞	22.31	25.03	14.77	59.26	71.81	63.17	63.70
	ω B97XD	1064	18.60	19.46	17.41	57.25	52.72	85.09	61.46
		1341	18.16	18.95	16.99	55.99	51.61	82.48	59.94
		1906	17.79	18.53	16.64	54.94	50.69	80.34	58.70
		∞	23.64	25.86	15.96	63.95	73.47	65.56	66.78

Table S9: (Continue)

Model	Method	λ	γ_{xxyy}	γ_{yyzz}	γ_{xxzz}	γ_{xxxx}	γ_{yyyy}	γ_{zzzz}	γ
SMD	B3LYP	1064	25.86	28.43	24.85	81.00	70.68	152.90	92.95
		1341	25.06	27.37	24.02	78.65	68.76	145.24	89.55
		1906	24.41	26.52	23.35	76.72	67.19	139.22	86.82
		∞	34.01	39.51	21.73	88.89	106.18	108.20	98.75
	CAM-B3LYP	1064	19.22	19.64	18.52	59.36	53.26	91.21	64.00
		1341	18.74	19.10	18.04	57.96	52.08	88.28	62.33
		1906	18.34	18.65	17.65	56.81	51.10	85.90	60.96
		∞	24.97	27.32	16.37	65.74	75.11	70.14	69.66
	M06-2X	1064	19.20	19.66	18.18	58.78	51.94	88.93	63.01
		1341	18.70	19.09	17.70	57.36	50.75	85.98	61.31
		1906	18.29	18.62	17.30	56.18	49.77	83.58	59.92
		∞	24.77	26.96	15.97	64.92	75.14	68.43	68.78
	M06-2X/ def2-TZVPPD	1064	17.62	18.74	17.63	55.22	49.40	84.00	59.64
		1341	17.18	18.22	17.17	53.96	48.35	81.28	58.11
		1906	16.82	17.80	16.80	52.91	47.47	79.07	56.86
		∞	23.62	25.92	15.49	61.38	74.20	66.39	66.41
	ω B97XD	1064	18.96	19.20	18.13	58.74	53.07	85.00	62.15
		1341	18.50	18.68	17.68	57.42	51.94	82.32	60.59
		1906	18.12	18.26	17.31	56.33	51.01	80.14	59.31
		∞	24.56	26.44	16.28	65.86	74.47	65.45	68.07

Table S10: Selected tensor components and field induced second-harmonic generation values, $\gamma_{\text{ESHG}}(-2\omega; \omega, \omega, 0) \times 10^{-36}$ esu, of OPPh₃py calculated at different levels of theory. All calculations employed the 6-311++G(2df,p) basis set unless otherwise indicated.

Model	Method	λ	γ_{xxyy}	γ_{yyzz}	γ_{xxzz}	γ_{xxxx}	γ_{yyyy}	γ_{zzzz}	γ_{ESHG}
GAS	B3LYP	1064	17.99	19.48	24.13	56.87	46.99	157.38	76.66
		1341	16.13	17.10	20.78	51.57	42.92	124.19	65.23
		1906	14.85	15.57	18.66	47.74	39.96	105.37	58.20
	CAM-B3LYP	1064	12.87	13.22	14.89	40.74	34.79	74.18	46.24
		1341	11.90	12.21	13.67	37.83	32.45	66.14	42.35
		1906	11.17	11.46	12.77	35.61	30.66	60.46	39.48
	M06-2X	1064	12.64	12.82	14.66	39.82	33.68	72.50	45.16
		1341	11.65	11.81	13.41	36.87	31.32	64.29	41.20
		1906	10.91	11.07	12.48	34.63	29.53	58.55	38.31
	M06-2X/ def2-TZVPPD	1064	10.74	11.75	13.08	35.10	29.40	65.73	40.18
		1341	9.97	10.88	12.02	32.68	27.55	58.41	36.82
		1906	9.39	10.23	11.24	30.83	26.12	53.31	34.37
	ω B97XD	1064	12.34	12.48	14.06	38.81	33.84	70.06	44.01
		1341	11.46	11.58	12.96	36.17	31.68	62.89	40.50
		1906	10.79	10.90	12.14	34.14	30.01	57.76	37.90
PCM	B3LYP	1064	28.24	27.57	34.00	86.17	75.52	196.92	107.15
		1341	25.36	24.59	29.84	78.15	68.86	160.73	93.14
		1906	23.33	22.55	27.05	72.28	63.98	138.83	83.97
	CAM-B3LYP	1064	20.27	19.55	22.04	61.53	55.27	101.01	68.05
		1341	18.66	18.04	20.22	57.01	51.39	90.36	62.34
		1906	17.46	16.91	18.87	53.56	48.43	82.76	58.11
	M06-2X	1064	20.20	19.10	22.05	60.85	54.24	98.75	67.07
		1341	18.55	17.57	20.15	56.24	50.30	87.91	61.23
		1906	17.32	16.44	18.75	52.74	47.30	80.24	56.93
	M06-2X/ def2-TZVPPD	1064	18.11	18.28	20.65	56.23	50.33	92.38	62.33
		1341	16.73	16.88	18.96	52.20	46.94	82.45	57.15
		1906	15.70	15.84	17.70	49.12	44.35	75.43	53.33
	ω B97XD	1064	19.74	18.94	21.51	60.14	54.88	96.13	66.08
		1341	18.25	17.56	19.83	55.94	51.22	86.41	60.80
		1906	17.13	16.51	18.57	52.71	48.41	79.42	56.86

Table S10: (Continue)

Model	Method	λ	γ_{xxyy}	γ_{yyzz}	γ_{xxzz}	γ_{xxxx}	γ_{yyyy}	γ_{zzzz}	γ_{ESHG}
SMD	B3LYP	1064	28.67	28.84	34.15	88.59	76.14	199.57	108.95
		1341	25.72	25.68	29.96	80.21	69.40	163.56	94.80
		1906	23.63	23.51	27.14	74.10	64.46	141.57	85.48
	CAM-B3LYP	1064	20.50	20.42	21.96	62.77	55.70	103.76	69.29
		1341	18.86	18.79	20.13	58.09	51.76	92.78	63.42
		1906	17.63	17.57	18.77	54.52	48.75	84.94	59.07
	M06-2X	1064	20.53	20.07	22.02	62.25	54.42	101.68	68.44
		1341	18.83	18.42	20.11	57.46	50.45	90.51	62.42
		1906	17.56	17.20	18.70	53.84	47.43	82.60	58.00
	M06-2X/ def2-TZVPPD	1064	18.50	19.31	20.79	57.56	50.74	94.98	63.77
		1341	17.06	17.78	19.06	53.37	47.30	84.76	58.41
		1906	15.99	16.64	17.78	50.18	44.67	77.52	54.47
	ω B97XD	1064	20.05	19.79	21.28	61.59	55.06	96.09	66.72
		1341	18.51	18.30	19.58	57.21	51.35	86.19	61.31
		1906	17.35	17.17	18.31	53.86	48.49	79.07	57.27

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